

The toroidal source potential, the uniform inducing field, and the GLOBAL1D↔MODEM normalization factor

Anna Kelbert with Claude

July 2026

Abstract

GLOBAL1D (`field1d.f90/field1d.s2.f90`) drives its solution with an *external toroidal potential amplitude* $\text{coeff}(l, m)$; MODEM’s regional 3-D solver drives its solution with a *boundary electric field* of unit amplitude. Because a *potential* and a *field* are related by Faraday’s law $\nabla \times \mathbf{E} = i\omega\mu_0\mathbf{H}$, the raw electric fields the two codes produce for their respective unit sources differ by a single multiplicative complex constant per period, $c = E_{\text{global1d}}/E_{\text{ModEM}} = i\omega\mu_0 G$, with G a real geometric constant set by the *air-layer thickness* and the surface magnetic response. This note explains the physics (the toroidal potential defines a quasi-uniform *magnetic* field, not an electric one), derives G from the air-layer geometry, and defines a normalized output preset that matches MODEM’s normalization. There is **no bug and no sign/time-convention error**: c cancels identically in the impedance $Z = E/H$, so the two codes agree on Z , apparent resistivity and transfer functions to $< 1\%$. The apparent “ $-\text{Re}, +\text{Im}$ vs $+\text{Re}, +\text{Im}$ sign flip” in raw fields is simply the $\approx 90^\circ$ phase of the $i\omega\mu_0$ factor.

1 The toroidal potential and the fields it produces

Under the quasi-static approximation a divergence-free field on a sphere admits a Mie (poloidal–toroidal) decomposition. In the insulating air there are no currents, so the poloidal electric potential P vanishes (Sun & Egbert 2012, eqs. 21–22) and an external MT/ionospheric source drives only the toroidal potential T . With $P \equiv 0$, eqs. (5)–(6) reduce to

$$\mathbf{E} = -i\omega\mu_0 \frac{\hat{\mathbf{r}}}{r} \times \nabla_a T, \quad \mathbf{H} = \frac{1}{r} \nabla_a \frac{\partial T}{\partial r} - \frac{\hat{\mathbf{r}}}{r^2} \nabla_a^2 T. \quad (1)$$

Writing $T(r, \theta, \phi) = \sum_{l,m} \text{coeff}(l, m) T_l(r) Y_l^m(\theta, \phi)$ and using $\hat{\mathbf{r}} \times \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}}$, $\nabla_a^2 Y_l^m = -l(l+1)Y_l^m$, the components are (`field1d.s2.f90`, l. 42–47)

$$\begin{aligned} E_\theta &= +\frac{i\omega\mu_0}{r} T_l Y_p, & E_\phi &= -\frac{i\omega\mu_0}{r} T_l Y_t, & E_r &= 0, \\ H_r &= +\frac{l(l+1)}{r^2} T_l Y, & H_\theta &= +\frac{1}{r} T_l' Y_t, & H_\phi &= +\frac{1}{r} T_l' Y_p, \end{aligned} \quad (2)$$

with $Y_t = \partial Y / \partial \theta$, $Y_p = (1/\sin \theta) \partial Y / \partial \phi$. The single scalar T generates a **toroidal electric field** ($E_r = 0$; the name refers to this) and a **poloidal magnetic field** ($H_r \neq 0$, the vertical/ Z component at the ground)—the standard deep-Earth induction picture.

2 The (external) toroidal potential is a quasi-uniform magnetic field

In the air T is harmonic radially, $T_l(r) = \alpha r^{l+1} + \beta r^{-l}$: an external (α , source) plus internal (β , induced) part. GLOBAL1D sets $\text{coeff}(l, m) = 1 \Rightarrow \alpha = 1$ (unit coefficient of r^{l+1} ; `field1d.s2.f90`,

l. 437–458). For $l = 1, m = 0$ with $Y_1^0 = \sqrt{3/4\pi} \cos \theta$ and $T_1^{\text{ext}} = \alpha r^2$,

$$H_r = 2\alpha \sqrt{\frac{3}{4\pi}} \cos \theta, \quad H_\theta = -2\alpha \sqrt{\frac{3}{4\pi}} \sin \theta, \quad (3)$$

and since $\cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\boldsymbol{\theta}} = \hat{\mathbf{z}}$,

$$\boxed{\mathbf{H}^{\text{ext}} = 2\alpha \sqrt{\frac{3}{4\pi}} \hat{\mathbf{z}}} \quad \text{a spatially uniform axial magnetic field.} \quad (4)$$

The induced electric field is *not* uniform and carries the explicit $i\omega\mu_0$: $E_\phi = +i\omega\mu_0 \alpha r \sqrt{3/4\pi} \sin \theta$. **Thus the (external, $l = 1$) toroidal potential defines a quasi-uniform inducing *magnetic* field, and E is the $i\omega\mu_0$ -scaled induced response.**

3 global1d vs ModEM: $c = i\omega\mu_0 G$

GLOBAL1D's unit source ($\propto$ uniform H) and ModEM's unit source (a boundary E -field) differ by the potential-vs-field relationship. The measured ratio $c = E_{\text{global1d}}/E_{\text{ModEM}}$ is (i) *constant over depth*—a multiplicative, not antilinear, relationship; (ii) *scaling as ω* : $|c|/\omega$ constant and $\arg c \rightarrow 90^\circ$ at short period $\Rightarrow c = i\omega\mu_0 G$; (iii) *mode-independent*. The i is Faraday's law; the small residual ($\arg c$ drifting $90^\circ \rightarrow 95^\circ$ over $T = 10 \rightarrow 1000$ s) is the earth's induction response (§4) plus ModEM's flat-vs-spherical boundary approximation.

4 Derivation of G from the air-layer geometry

ModEM's surface field. ModEM drives a unit E at the top of the domain, a height $d_{\text{air}} \approx 1000$ km above the surface, and solves a 1-D column (Mod2d/WSfwd1Dmod.f90). In the insulating air $d^2 E/dz^2 = 0$, so E is linear in height and H is uniform (as in §2). With $E(0) = 1$ and surface impedance $Z_e = E_s/H_s$,

$$E_{\text{ModEM}}(\text{surface}) = \frac{Z_e}{Z_e - i\omega\mu_0 d_{\text{air}}} \approx \frac{-Z_e}{i\omega\mu_0 d_{\text{air}}}, \quad (5)$$

since $|Z_e| \ll \omega\mu_0 d_{\text{air}}$ for a 1000 km air column. ModEM's surface field is set by the air thickness.

The ratio. GLOBAL1D obeys $E_\phi = Z_e H_\theta$ at the surface (same Z_e). Taking the ratio, Z_e cancels:

$$c = \frac{E_{\text{global1d}}}{E_{\text{ModEM}}} = H_\theta (Z_e - i\omega\mu_0 d_{\text{air}}) = E_\phi - i\omega\mu_0 d_{\text{air}} H_\theta = -i\omega\mu_0 H_\theta (d_{\text{air}} + C), \quad (6)$$

where $C \equiv Z_e/(i\omega\mu_0)$ is the C-response ($\approx 4, 28, 132$ km at $T = 10, 100, 1000$ s). Hence

$$\boxed{G = -H_\theta(\text{surface}) (d_{\text{air}} + C) \approx -H_\theta d_{\text{air}}} \quad (7)$$

the surface horizontal field times the air-layer thickness; C is the period-dependent residual.

The geometric value of H_θ . From $T_1(r) = r^2 + \beta/r$ at r_0 , $T_1'(r_0) = 2r_0 - \beta/r_0^2$, so

$$H_\theta(\text{surface}) = -\left(2 - \frac{\beta}{r_0^3}\right) \sqrt{\frac{3}{4\pi}} \sin \theta = -(2 - Q_1) \sqrt{\frac{3}{4\pi}} \sin \theta, \quad (8)$$

with $Q_1 = \beta/r_0^3$ the $l = 1$ Q-response (limits $Q_1 = 0$ insulator, $Q_1 = l/(l+1) = \frac{1}{2}$ perfect conductor). At MT/geomagnetic periods the mantle is a good conductor, $Q_1 \rightarrow \frac{1}{2}$, and at the equator

$$G = \frac{3}{2} \sqrt{\frac{3}{4\pi}} d_{\text{air}} \approx 0.733 d_{\text{air}} \quad (9)$$

Here $\frac{3}{2} = (2 - Q_1)$ combines external (2) and induced ($-Q_1$) parts, $\sqrt{3/4\pi}$ is the Y_1^0 amplitude, and $d_{\text{air}} = \text{grid}\%r(1) \cdot 1000 - r_0 \approx 1.0 \times 10^6$ m.

Numerical verification.

T (s)	$ H_\theta $	$ G /d_{\text{air}}$ (meas.)	$\arg G$
10	0.718	0.719	+0.65°
100	0.721	0.730	+1.59°
1000	0.720	0.783	+4.73°

The derived $\frac{3}{2} \sqrt{3/4\pi} = 0.733$ straddles the measured $|G|/d_{\text{air}}$ (best at 100 s); the drift and growing $\arg G$ are the $(d_{\text{air}} + C)$ correction. H_θ is nearly period-independent, confirming it is geometric in the deep-induction limit ($Q_1 \approx \frac{1}{2}$).

5 The normalized output preset

To express GLOBAL1D's fields in MODEM's normalization, multiply by $1/c$:

$$(\mathbf{E}, \mathbf{H})_{\text{norm}} = \frac{1}{i\omega\mu_0 G} (\mathbf{E}, \mathbf{H}), \quad G = \frac{3}{2} \sqrt{\frac{3}{4\pi}} d_{\text{air}}, \quad (10)$$

with d_{air} from the grid and ω from the period. Applying the same complex factor to E and H (equivalently rotating coeff by $1/i = -i$ and scaling) **preserves** $Z = E/H$: the physics is untouched; only the source-normalization convention changes. Implemented as the `EGBERTKELBERT2012.MODEM` preset (`output_convention.f90`). The factor removes the exact $i\omega\mu_0$ and the air-thickness scaling; residuals (the C -response, finite Q , and flat-vs-spherical) leave a few-percent amplitude and $\lesssim 5^\circ$ phase mismatch at long period, $\rightarrow 0$ at short period—genuine physics differences, not artifacts.

6 Validation against Cartesian ModEM

The spherical MODEM runs used to *measure* c (`Mod3DMT_SP2_SPH`) require building with `-DBUILD_SPHERICAL -DFORCE_SPHERICAL`, bypassing a guard that warns the spherical version is *experimental and “will not work with traditional MT source setup”*—and the internal `COMPUTE_BC` boundary condition IS that traditional (WS/REBOCC per-column) setup. The derivation was therefore cross-checked against **Cartesian ModEM** (trusted and accurate) on a $\rho=100\Omega\text{m}$ halfspace with the same 1000 km/12-layer air.

Air-column mechanism (the heart of G): Cartesian MODEM’s surface E equals the analytic $Z_e/(Z_e - i\omega\mu_0 d_{\text{air}})$ (with $Z_e = -i\omega\mu_0/\sqrt{-i\omega\mu_0\sigma}$) to 0.2–0.6%, $\pm 0.1^\circ$ at every period from 4 to 4000 s.

Full preset, $r = E_{\text{global1d}}^{\text{norm}}(\text{equator})/E_{\text{MODEM}}^{\text{cart}}$:

T (s)	3.98	1000	3981
$ r $	1.006	1.073	1.145
$\arg r$	$+0.2^\circ$	$+3.6^\circ$	$+6.4^\circ$

The match is essentially perfect in the flat-earth limit ($|r| = 1.006$, $+0.2^\circ$ at 4 s, where G is exact and $Q_1 \rightarrow \frac{1}{2}$) and the residual grows monotonically and cleanly with period as curvature enters; Cartesian MODEM is laterally uniform to 0.00%. This validates the $i\omega\mu_0 G$ framework on trusted code and shows the residual is the **genuine flat-vs-spherical** physics—GLOBAL1D being the more accurate spherical solution—not an artifact of the experimental spherical BC.

Generality. The derivation is for $l = 1$ (P10 / MT modes) at the equator in the deep-induction limit. In general $H_\theta = -[(l+1) - l Q_l] r_0^{l-1} \partial_\theta Y_l^0$ depends on (l, θ) , so a single scalar G is exact only for the $l = 1$ equatorial case the MT modes reduce to—the relevant limit for MT interoperability.